DATA ANALYTICS CHALLENGE BY ANNIE

Official Challenge Guidelines and Instructions

Welcome to the Data Analytics Challenge by Annie!

This challenge is designed to test your practical ability to work as a data analyst, from raw data to meaningful business insights. It is an opportunity to demonstrate your analytical thinking, technical skills, problem-solving ability, and capacity to communicate data-driven recommendations.

This is not just a dashboard competition. You are expected to demonstrate the complete data analytics process, including data cleaning, exploratory data analysis, business problem definition, analysis, visualization, documentation, and recommendations.

IMPORTANT: AI USE IS STRICTLY PROHIBITED.

Your work must reflect your own knowledge, reasoning, and technical abilities. Read all instructions carefully before beginning.

Challenge Details

Challenge name	Data Analytics Challenge by Annie
Dataset	FIFA Football Data
Challenge focus	End-to-end data analytics
Submission link	To be announced
Submission deadline	To be announced
Submission format	Project files, analytical documentation, visualizations, and final presentation of findings, as applicable

1. Challenge Overview

The objective is to evaluate how well you can transform raw football data into useful insights that answer relevant questions and support informed decisions. Approach the challenge as a professional data analyst working with a real-world dataset.

2. About the Dataset

The dataset provided for this challenge contains FIFA football-related data. Depending on the specific dataset version, it may include information about football players, their attributes, positions, clubs, nationalities, ages, ratings, potential, market values, wages, physical characteristics, or other player-related statistics. The exact columns and available variables depend on the dataset provided.

Your first responsibility is to understand the data before attempting to analyze it. Investigate:

- What does each row represent?
- What does each column mean?
- What types of data are available?
- Which variables are numeric, categorical, or date-related?
- Are there missing, duplicated, inconsistent, or incorrectly formatted values?
- What questions can this dataset reasonably answer?
- What limitations might affect your conclusions?

Your goal is not simply to describe football statistics. Your goal is to discover meaningful patterns, explain what they mean, and communicate your findings clearly.

3. Challenge Objectives

By the end of this challenge, you should demonstrate that you can:

- Understand and assess an unfamiliar dataset.
- Clean and prepare data for reliable analysis.
- Define relevant business or analytical questions.
- Perform exploratory data analysis.
- Apply appropriate analytical techniques.
- Identify patterns, trends, relationships, and outliers.
- Develop clear, effective data visualizations.
- Extract meaningful insights from your findings.
- Provide evidence-based recommendations.
- Document your analytical process and decisions.
- Communicate findings to technical and non-technical audiences.

You are assessed on the quality of your reasoning and analysis, not merely on how attractive your dashboard looks.

4. Required Project Tasks

Task 1: Understand the Dataset

Before cleaning or analyzing the data, conduct an initial assessment. Include the number of rows and columns; a description of the dataset and its purpose; the meaning of important variables; data types; initial data-quality observations; potential areas of analysis; and any assumptions. Be able to explain what information the dataset contains and what it can help you investigate.

Expected output: A brief dataset overview and initial data assessment.

Task 2: Data Cleaning and Preparation

Identify and address relevant data quality issues. Consider missing or null values, duplicate records, incorrect data types, inconsistent categories, text formatting, unnecessary columns, invalid numeric values, currency or measurement inconsistencies, extra spaces, outliers, and other issues. For each significant cleaning decision, document what you found, what action you took, and why. Do not delete records or replace missing values without considering the implications. An unusually high player valuation, for example, may be legitimate rather than an error. Preserve the original dataset and work on a separate copy.

Expected output: A cleaned dataset and a record of cleaning steps, decisions, and assumptions.

Task 3: Define Business or Analytical Questions

Define specific, relevant questions that can be answered using the available data. Possible areas include player performance and overall ratings; potential compared with current ability; age and performance relationships; differences between positions; club or national team comparisons; market value and wages; nationality representation; physical attributes and ratings; or identifying potentially undervalued players. These are examples, not questions you must copy. For each question, explain what you want to investigate, why it is relevant, which variables are needed, how you intend to analyze it, and what insight the answer could provide. Do not invent a problem that the data cannot support. If you frame your work around scouting, recruitment, valuation, or team planning, distinguish analytical applications from established facts about the dataset.

Expected output: A documented list of analytical questions and the rationale behind them.

Task 4: Exploratory Data Analysis (EDA)

Explore the dataset using appropriate descriptive statistics and analytical techniques. These may include mean, median, minimum and maximum; frequency distributions; counts and percentages; group comparisons; distribution analysis; correlation analysis where appropriate; trend analysis if a meaningful time dimension exists; outlier investigation; and segmentation by position, age group, nationality, club, or other relevant variables. Let your questions guide the exploration. Avoid producing statistics merely to fill a report.

Expected output: Documented exploratory analysis with relevant tables, calculations, and observations.

Task 5: Perform Your Analysis

Use your exploration to answer the questions you selected. Demonstrate that you can select appropriate methods, perform accurate calculations, interpret results, and explain your reasoning. For each major question, state the question, explain the method, present results, identify the finding, explain why it matters, and mention limitations or alternative explanations. Correlation does not automatically establish causation. Do not claim that one attribute causes another unless your methodology supports that conclusion.

Expected output: A structured analysis that answers your questions using evidence from the dataset.

Task 6: Data Visualization and Dashboard Development

Create appropriate visualizations using a suitable analytics or visualization tool that you can operate independently. Charts should match the question, have clear titles and labels, use readable scales and legends, display units correctly, avoid misleading representations, and communicate patterns without unnecessary decoration. Possible types include bar charts, histograms, scatter plots, box plots, line charts, heatmaps, and others suited to the data. If you build a dashboard, organize it logically and include relevant filters or interactive features where useful. Every visualization should serve a clear analytical purpose.

Expected output: A coherent set of visualizations and, where appropriate, an interactive dashboard.

Task 7: Generate Insights and Recommendations

Explain what your findings mean. For each major finding, state what you discovered, what evidence supports it, why it matters, what decision or action it could inform, and what limitations should be considered. Where relevant, develop practical recommendations for a potential audience such as football scouts, recruitment analysts, club management, or sports researchers. For example, a finding that some players have high ratings relative to market value could support further scouting, but does not by itself prove that a player is a good investment. Consider factors the dataset may not contain, such as contract terms, injuries, transfer fees, and future performance. Recommendations must follow logically from your findings.

Expected output: Evidence-based insights and actionable recommendations.

Task 8: Document Your Work

Your documentation should allow another analyst to understand your process, review your decisions, and reproduce your work where possible. Include: project title; introduction and dataset overview; problem statement; objectives; analytical questions; tools used; data-cleaning process; methodology; findings supported by evidence and visuals; insights and recommendations; limitations; and conclusion. Include relevant screenshots, tables, charts, formulas, SQL queries, notebooks, or other supporting materials. Write the documentation in your own words.

Expected output: A complete project report or equivalent analytical documentation.

Task 9: Organize Your Project Files

Keep files organized for review. Where applicable, include the original dataset if permitted, cleaned dataset, analysis files such as spreadsheets, SQL scripts or notebooks, dashboard or visualization file, project documentation, and supporting charts, tables, and screenshots. Use meaningful file names and ensure files open correctly. If a dashboard requires a particular application, state the application and include the relevant project file. Do not submit only screenshots if the underlying work can reasonably be provided.

Expected output: A clearly organized project folder. Specific submission instructions and the submission link will be announced separately.

5. Tools and Independent Work

You may use data analytics tools and software you already know, provided they do not use AI to perform any part of the challenge. Depending on your approach, these may include:

- Microsoft Excel
- SQL
- Python
- Power BI

- Tableau
- Other conventional, non-AI analytical tools

You do not need to use every tool listed. Choose tools appropriate to your approach and demonstrate that you understand the work performed in them. You are responsible for ensuring that your tools and their features comply with the no-AI rule.

6. Strict No-AI Policy

AI USE IS STRICTLY PROHIBITED THROUGHOUT THIS CHALLENGE.

The purpose of this challenge is to evaluate your own data analytics skills, critical thinking, problem-solving ability, and technical competence. You must complete the challenge independently, using your own knowledge and skills.

The following are prohibited:

- Using ChatGPT, Claude, Gemini, Copilot, or other generative AI assistants to complete any part of the challenge.
- Using AI to clean, transform, analyze, or interpret the dataset.
- Generating SQL queries, Python code, Excel formulas, DAX measures, or other analytical solutions using AI.
- Using AI to generate business questions, insights, conclusions, or recommendations for your submission.
- Using AI to create reports, documentation, presentation content, or dashboard designs for this challenge.
- Using AI-powered features that automatically perform analytical tasks on your behalf.
- Submitting AI-generated work as your own.
- Asking another person or service to complete any part of the challenge for you.

You may consult ordinary software documentation, provided you do not use AI assistance to produce your solution. Standard, non-AI software features are permitted when they do not violate the policy. Examples include conventional spreadsheet formulas, SQL queries you write yourself, standard Power Query transformations, and manually created visualizations. If you are unsure whether a feature is AI-powered, do not use it unless you receive clarification from the challenge organizer.

Consequences of Violating the Policy

Any participant found to have used AI or submitted work produced with prohibited assistance may be disqualified. This rule applies even if the violation is discovered after the participant has been announced as a winner or has already received a prize or other benefit.

Where a violation is established, the organizer reserves the right to:

- Disqualify the participant.
- Revoke any award, prize, certificate, recognition, or other benefits received through the challenge.
- Require the participant to refund any monetary prize or payment already received as a result of winning.
- Require the forfeiture of any remaining benefits attached to the award.

These consequences apply in accordance with the challenge rules and any applicable legal requirements. Participants may be asked to explain their methodology, demonstrate their files, walk through calculations, or reproduce parts of their analysis to verify that they understand and performed the work themselves. Reviewers may examine the consistency of submitted work and request supporting evidence. No single indicator should automatically be treated as conclusive proof of AI use without appropriate review.

By participating in this challenge, you agree to comply with this policy.

7. Assessment Criteria

Submissions will be evaluated using the following scoring framework:

Assessment area	Weight
Dataset understanding and initial assessment	5%

Assessment area	Weight
Data cleaning and preparation	15%
Business problem definition and analytical questions	10%
Exploratory data analysis	10%
Analytical methods, accuracy, and reasoning	20%
Data visualization and dashboard quality	15%
Insights and actionable recommendations	10%
Documentation, reproducibility, and file organization	10%
Communication and presentation of findings	5%
TOTAL	100%

What Reviewers Will Look For

Dataset understanding (5%) Do you understand the structure, variables, and limitations of the dataset? Can you explain what the data represents?

Data cleaning (15%) Did you identify relevant data-quality issues? Were your cleaning decisions appropriate, documented, and justified?

Business questions (10%) Are your questions clear, relevant, specific, and answerable using the available data?

Exploratory analysis (10%) Did you investigate the data meaningfully using appropriate descriptive statistics and exploratory techniques?

Analytical accuracy and reasoning (20%) Are your calculations correct? Are your methods appropriate? Can you explain how you reached your conclusions and why your approach makes sense?

Visualization (15%) Are your charts and dashboard clear, accurate, readable, and appropriate for the questions being answered?

Insights and recommendations (10%) Do your findings go beyond describing numbers? Are your recommendations practical, relevant, and supported by evidence?

Documentation and reproducibility (10%) Is your work properly documented? Can another analyst follow your process and inspect the supporting files?

Communication (5%) Can you communicate the most important findings clearly and concisely to someone without a technical background?

A visually attractive dashboard will not compensate for inaccurate analysis, poor data cleaning, weak reasoning, or unsupported conclusions. Likewise, a technically sophisticated project will not receive full credit if its findings are poorly communicated or its methodology is undocumented.

8. Submission Requirements

The official submission link will be provided and announced separately. Before submitting, ensure that you have:

- Completed all relevant stages of the analysis.
- Answered your stated analytical questions.
- Checked the accuracy of your calculations.
- Documented your cleaning decisions and methodology.
- Included your visualizations and key findings.
- Provided practical, evidence-based recommendations.
- Organized your project files clearly.
- Confirmed that your files open and work correctly.

- Ensured that every part of the submission complies with the no-AI policy.

Please do not submit through unofficial channels unless instructed to do so.

Submission link: To be announced

Submission deadline: To be announced

Further submission details will be communicated by the organizer.

9. Final Advice to Participants

This challenge is not about producing the most complicated dashboard or using the largest number of tools. It is about demonstrating sound analytical thinking. Take time to understand the data before jumping into visualization. Ask meaningful questions. Clean the data carefully. Validate your calculations. Explain your findings. Be honest about limitations.

Do not force conclusions that the data cannot support. Most importantly, make sure you can explain every significant decision you make, from how you handled missing values to why you selected a particular chart or recommendation. Your ability to think critically, work independently, and defend your analytical decisions is what matters.

FINAL REMINDER

This is the Data Analytics Challenge by Annie. The work you submit must be your own. No AI assistance is permitted.

Approach the challenge with curiosity, discipline, and integrity. Let your analysis speak for itself.